PERFORMANCE SPECIFICATION SHEET
CAPACITORS, CHIP, FIXED, TANTALUM, POLARIZED
ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY,
STYLE CWR16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55365.

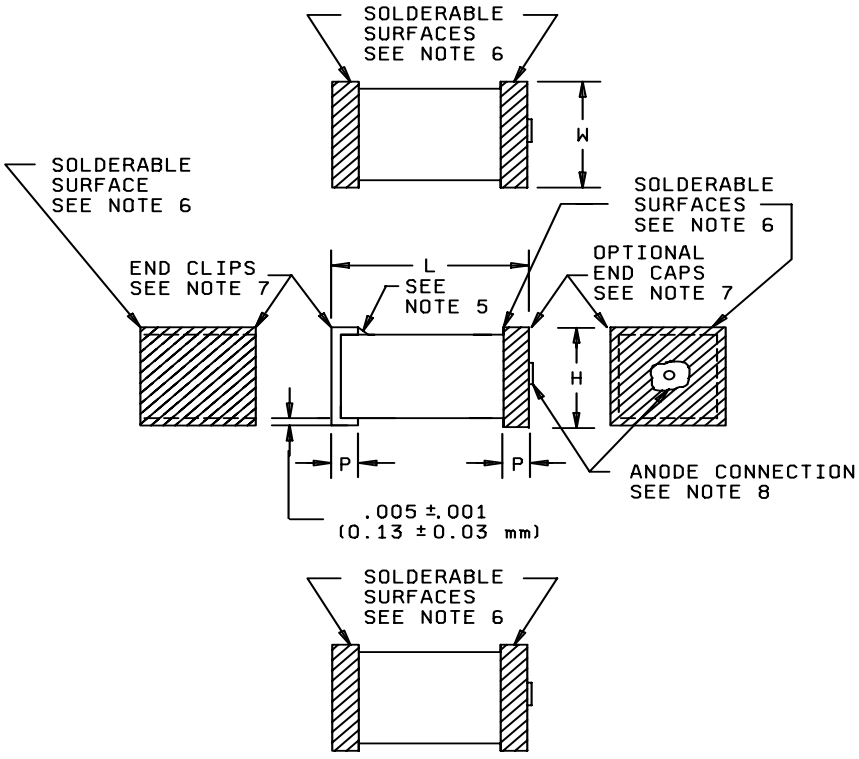


FIGURE 1. Style CWR16 [conformally coated] capacitors.

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Case size	L	W	H	P
	± .015 (0.38)	±.015 (0.38)	±.015 (0.38)	±.005 (0.13)
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. Cathode terminal may be epoxy coated on one surface for polarity identification.
6. Solderable surfaces are only those surfaces designated as such.
7. Anode and cathode terminals may be two-sided through five-sided, and may not be the same on some designs, but will meet all specification requirements.
8. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 (0.38 mm) maximum. The riser wire weld area is not solderable. The riser wire connection shall have a continuous weld of a minimum of 25 percent of the contact surface.
9. When solder coated terminations are required, add .015 (0.38 mm) to termination dimension tolerances.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Termination finish: In accordance with termination finishes B, C, H, or K of MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I.

Capacitance: See table I.

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Capacitance tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 (see table I).

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Step 1 (+25°C):	DCL	- See table I.
	Capacitance	- Shall not exceed tolerance of table I value.
	DF	- See table I.
Step 2 (-55°C):	Δ Capacitance	- Shall not exceed ±10 percent of initial measurement.
	DF	- See table I.
Step 3 (+25°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±5 percent of initial +25°C measurement.
	DF	- See table I.
Step 4 (+85°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±10 percent of initial +25°C measurement.
	DF	- See table I.
Step 5 (+125°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±15 percent of initial +25°C measurement.
	DF	- See table I.
Step 6 (+25°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±5 percent of initial +25°C measurement.
	DF	- See table I.

Surge voltage: In accordance with MIL-PRF-55365.

DCL	- See table I.
Δ Capacitance	- Within ±5 percent of initial value.
DF	- See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C:	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±5 percent of initial measured value.
	DF	- See table I.
At 85°C:	DCL	- See table I.
At 125°C:	DCL	- See table I.

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10,000 hours:

At 25°C:	DCL	- 200 percent of initial requirement.
	∆ Capacitance	- Shall not exceed ±5 percent of initial measured value.
	DF	- See table I.
At 85°C:	DCL	- 200 percent of initial requirement
At 125°C:	DCL	- 200 percent of initial requirement.

Solderability: In accordance with MIL-PRF-55365 except that following steam aging, test samples may have a 30 minute bake out at 150°C prior to solder dipping and all terminations shall exhibit a continuous solder coating free from defects for a minimum of 75 percent of the critical area of any individual termination.

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TABLE I. Style CWR16 characteristics.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (µF)	DC Leakage (max) (µA)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR16C-335--A-	4	3.3	1	10	12	6	8	8	12	A
CWR16C-475--A-	4	4.7	1	10	12	6	8	8	12	A
CWR16C-685--A-	4	6.8	1	10	12	6	8	8	12	A
CWR16C-106--B-	4	10	1	10	12	8	10	10	8	B
CWR16C-156--B-	4	15	1	10	12	8	10	10	8	B
CWR16C-226--B-	4	22	1	10	12	8	10	10	8	B
CWR16C-226--D-	4	22	1	10	12	8	10	12	4	D
CWR16C-336--D-	4	33	2	20	24	8	10	12	4	D
CWR16C-336--E-	4	33	2	20	24	8	10	12	3	E
CWR16C-476--E-	4	47	2	20	24	8	10	12	3	E
CWR16C-686--E-	4	68	3	30	36	8	10	12	3	E
CWR16C-107--F-	4	100	4	40	48	10	12	12	2	F
CWR16C-157--G-	4	150	6	60	72	10	12	12	1	G
CWR16C-227--H-	4	220	8	80	96	10	12	12	1	H
CWR16C-337--H-	4	330	10	100	120	10	12	12	0.9	H
CWR16D-335--A-	6	3.3	1	10	12	6	8	8	12	A
CWR16D-475--A-	6	4.7	1	10	12	6	8	8	12	A
CWR16D-685--B-	6	6.8	1	10	12	6	8	8	8	B
CWR16D-106--B-	6	10	1	10	12	6	8	8	8	B
CWR16D-156--B-	6	15	1	10	12	8	10	10	8	B
CWR16D-156--D-	6	15	1	10	12	8	10	12	5	D
CWR16D-156--E-	6	15	1	10	12	8	10	12	3	E
CWR16D-226--D-	6	22	1	10	12	6	8	8	5	D
CWR16D-226--E-	6	22	2	20	24	8	10	12	3.5	E
CWR16D-336--E-	6	33	2	20	24	6	8	8	3.5	E
CWR16D-476--F-	6	47	3	30	36	8	10	12	3.5	F
CWR16D-686--F-	6	68	4	40	48	10	12	12	1.5	F
CWR16D-686--G-	6	68	4	40	48	10	12	12	1	G
CWR16D-107--G-	6	100	6	60	72	10	12	12	1.1	G
CWR16D-157--G-	6	150	10	100	120	10	12	12	1.1	G
CWR16D-227--H-	6	220	10	100	120	10	12	12	0.9	H
CWR16D-337--H-	6	330	20	200	240	10	12	12	0.9	H

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Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (µF)	DC Leakage (max) (µA)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR16F-225--A-	10	2.2	1	10	12	6	8	8	12	A
CWR16F-335--A-	10	3.3	1	10	12	6	8	8	12	A
CWR16F-475--B-	10	4.7	1	10	12	6	8	8	8	B
CWR16F-475--C-	10	4.7	1	10	12	6	8	8	5.5	C
CWR16F-685--B-	10	6.8	1	10	12	6	8	8	8	B
CWR16F-685--C-	10	6.8	1	10	12	6	8	8	5.5	C
CWR16F-685--D-	10	6.8	1	10	12	6	8	8	5	D
CWR16F-106--B-	10	10	1	10	12	8	10	10	8	B
CWR16F-106--C-	10	10	1	10	12	6	8	8	5.5	C
CWR16F-106--D-	10	10	1	10	12	6	8	8	4	D
CWR16F-106--E-	10	10	1	10	12	6	8	8	3.5	E
CWR16F-156--D-	10	15	1	10	12	6	8	8	5	D
CWR16F-156--E-	10	15	2	20	24	8	10	10	3	E
CWR16F-226--E-	10	22	3	30	36	8	10	10	2	E
CWR16F-336--F-	10	33	3	30	36	8	10	10	1.5	F
CWR16F-476--F-	10	47	4	40	48	10	12	12	1.5	F
CWR16F-476--G-	10	47	4	40	48	10	12	12	1	G
CWR16F-686--G-	10	68	6	60	72	10	12	12	1.1	G
CWR16F-107--G-	10	100	10	100	120	10	12	12	1.1	G
CWR16F-107--H-	10	100	10	100	120	10	12	12	0.9	H
CWR16F-157--H-	10	150	15	150	180	10	12	12	0.9	H
CWR16F-227--H-	10	220	20	200	240	10	12	12	0.9	H
CWR16H-105--A-	15	1	1	10	12	6	8	8	15	A
CWR16H-155--A-	15	1.5	1	10	12	6	8	8	15	A
CWR16H-225--A-	15	2.2	1	10	12	6	8	8	15	A
CWR16H-335--B-	15	3.3	1	10	12	6	8	8	9	B
CWR16H-475--B-	15	4.7	1	10	12	6	8	8	5	B
CWR16H-475--C-	15	4.7	1	10	12	6	8	8	5.5	C
CWR16H-475--D-	15	4.7	1	10	12	6	8	8	6	D
CWR16H-685--D-	15	6.8	1	10	12	6	8	8	6	D
CWR16H-685--E-	15	6.8	1	10	12	8	10	12	3	E
CWR16H-106--D-	15	10	2	20	24	6	8	8	6	D
CWR16H-106--E-	15	10	2	20	24	6	8	8	4	E
CWR16H-156--E-	15	15	2	20	24	6	8	8	4	E
CWR16H-156--F-	15	15	2	20	24	8	10	10	3	F

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Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (µF)	DC Leakage (max) (µA)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR16H-226--F-	15	22	3	30	36	8	10	10	3	F
CWR16H-336--F-	15	33	5	50	60	6	8	8	3	F
CWR16H-336--G-	15	33	6	60	72	8	10	10	1.1	G
CWR16H-476--G-	15	47	10	100	120	8	10	10	1.1	G
CWR16H-476--H-	15	47	10	100	120	8	10	10	0.9	H
CWR16H-686--G-	15	68	10	100	120	8	10	10	1.1	G
CWR16H-686--H-	15	68	10	100	120	8	10	10	0.9	H
CWR16H-107--H-	15	100	15	150	180	10	12	12	0.9	H
CWR16J-684--A-	20	0.68	1	10	12	6	8	8	15	A
CWR16J-105--A-	20	1	1	10	12	6	8	8	15	A
CWR16J-155--B-	20	1.5	1	10	12	6	8	8	9	B
CWR16J-225--B-	20	2.2	1	10	12	6	8	8	9	B
CWR16J-335--D-	20	3.3	1	10	12	6	8	8	6	D
CWR16J-475--E-	20	4.7	1	10	12	6	8	8	6	E
CWR16J-685--E-	20	6.8	2	20	24	6	8	8	5	E
CWR16J-106--E-	20	10	2	20	24	6	8	8	5	E
CWR16J-106--F-	20	10	2	20	24	6	8	8	3	F
CWR16J-156--F-	20	15	3	30	36	6	8	8	3	F
CWR16J-226--G-	20	22	4	40	48	8	10	10	2.5	G
CWR16J-336--H-	20	33	6	60	72	8	10	10	0.9	H
CWR16J-476--H-	20	47	10	100	120	8	10	10	0.9	H
CWR16K-474--A-	25	0.47	1	10	12	6	8	8	15	A
CWR16K-105--B-	25	1	1	10	12	6	8	8	10	B
CWR16K-225--D-	25	2.2	1	10	12	6	8	8	6	D
CWR16K-335--E-	25	3.3	1	10	12	6	8	8	4	E
CWR16K-685--F-	25	6.8	2	20	24	6	8	8	3	F
CWR16K-156--G-	25	15	4	40	48	6	8	8	1.4	G
CWR16K-226--G-	25	22	6	60	72	6	8	8	1.4	G
CWR16K-226--H-	25	22	6	60	72	6	8	8	0.9	H
CWR16K-336--H-	25	33	10	100	120	8	10	10	0.9	H
CWR16M-334--A-	35	0.33	1	10	12	6	8	8	22	A
CWR16M-685--G-	35	6.8	3	30	36	6	8	8	1.5	G
CWR16M-106--H-	35	10	4	40	48	8	10	10	0.9	H
CWR16M-156--H-	35	15	6	60	72	6	8	8	0.9	H

1/ Complete PIN shall include additional symbols to indicate capacitance tolerance, product level designator, and surge current option letter. If surge current is not required, the last "-" shall be replaced with the letter Z.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-2293)

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